



Signals and Systems EE252, Spring 2026

Instructor: Tariq Mumtaz

Semester week # 9 -10





Course Learning Outcome (CLOs):

CLO-1	Apply knowledge of signals and systems to categorize and perform basic operations on signals and systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-2	Calculate the response of DT and CT LTI systems to various inputs and categorize LTI systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-3	Analyze DT signals and systems in the transform domain using the Z transform.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-4	Analyze CT signals and systems in the transform domain using Laplace, Fourier transforms, and Fourier Series.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-5	Investigate complex engineering problems in signals and systems and formally present the results.	PLO-4 (Investigation)	Cognitive Level-5 (Evaluating)



Need for Transforms in Signal & Systems Analysis:

- Transforms provide frequency-domain insight
- Transforms help to attain a compact mathematical description of Signals & Systems using complex exponentials.
 - Transforms allow decomposition of the signal into complex sinusoids ($e^{j\omega}$)
 - Transforms also allow system representation in exponential form ($e^{\sigma+j\omega}$ or $re^{j\omega}$)
- Transforms convert convolution in the time domain into multiplication in the frequency domain
- Transform converts differential (continuous time)/ difference equations (discrete time) into algebraic equations.



Need for Transforms in Signal & Systems Analysis:

- CTFT/DTFT: decomposition into pure complex exponentials $e^{j\omega t}$ or $e^{j\Omega n}$.
- Laplace/Z-Transform: decomposition into general complex exponentials e^{st} or z^n
- CTFT and DTFT are primarily used for spectral (frequency) analysis, while Laplace and Z-transforms are more powerful for system analysis because they include stability/causality and transient behavior.

$$\text{CTFT} = X(s) \Big|_{s=j\omega}$$

$$x(t) \xleftrightarrow{\text{CTFT}} X(\omega)$$

$$x(t) \xleftrightarrow{\text{Laplace Transform}} X(s)$$

$$x[n] \xleftrightarrow{\text{DTFT}} X(e^{j\Omega})$$

$$x[n] \xleftrightarrow{\text{Z Transform}} X(z)$$

$$\text{DTFT} = X(z) \Big|_{z=e^{j\Omega}}$$



Roadmap till Semester end:

Frequency Domain Analysis of CT/DT Signals & Systems:

- ➔ Z- Transform (Wk 7 – 10)
- Laplace Transform (Wk 11 – 13)
- Fourier Transform (Wk 14 – 15)
- Project presentations (Wk16)



Learning outcome of week# 7 & 8 sessions:

Z- Transform – 5.1 & 5.2

- Z Transform – Signals Perspective
- Concept & Formulae
- Z Transform and Region of Convergence (ROC)
- Z Transform properties
- Z- Inverse and Partial fraction expansion



Learning outcome of today's session:

Z- Transform solution of Difference Equation – 5.3

- Z Transform – Systems Perspective
- DT System Modeling from Difference Equations
- Solution of Difference Equations:
 - Time- domain Recursive/iterative method
 - DT System total response – Zero input & Zero State response
 - Frequency-domain Z transform method



Z Transform: Signals Perspective



Z Transform:

$$X[z] = \sum_{n=0}^{\infty} x[n]z^{-n} = x[0] + \frac{x[1]}{z} + \frac{x[2]}{z^2} + \frac{x[3]}{z^3} + \dots$$

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

here z is a complex variable. The signal $x[n]$, which is the inverse z -transform of $X[z]$, can be obtained from $X[z]$ by using the following inverse z -transformation:

$$x[n] = \frac{1}{2\pi j} \oint X[z]z^{n-1} dz$$

That's the integration around a closed path in the complex plane.

Z Transform:

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$x[n]$ = discrete-time signal

z = complex variable

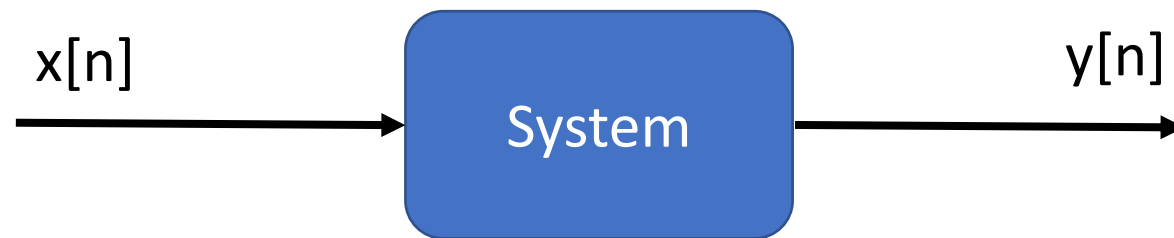
$$z = re^{j\omega}$$

$X(z)$ = Z-transform of the signal

$$X[z] = \mathcal{Z}\{x[n]\} \quad \text{and} \quad x[n] = \mathcal{Z}^{-1}\{X[z]\}$$

$$x[n] \Longleftrightarrow X[z]$$

Z Transform: Systems Perspective





Systems Perspective

Model DT LTI Systems as Difference Equations

DT LTI System Model:

Going back to chapter 03 – Discrete Time Signals & Systems.

3-4: EXAMPLES OF DISCRETE-TIME SYSTEMS

EXAMPLE 3.6 Savings Account

A person makes a deposit (the input) in a bank regularly at an interval of T (say, 1 month). The bank pays a certain interest on the account balance during the period T and mails out a periodic statement of the account balance (the output) to the depositor. Find the equation relating the output $y[n]$ (the balance) to the input $x[n]$ (the deposit).

In this case, the signals are inherently discrete time. Let

$x[n]$ = deposit made at the n th discrete instant
 $y[n]$ = account balance at the n th instant computed immediately after receipt of the n th deposit $x[n]$
 r = interest per dollar per period T

The balance $y[n]$ is the sum of (i) the previous balance $y[n-1]$, (ii) the interest on $y[n-1]$ during the period T , and (iii) the deposit $x[n]$

$$\begin{aligned} y[n] &= y[n-1] + ry[n-1] + x[n] \\ &= (1+r)y[n-1] + x[n] \end{aligned}$$

or

$$y[n] - ay[n-1] = x[n] \quad a = 1+r \quad (3.3)$$

The signals are inherently discrete-time

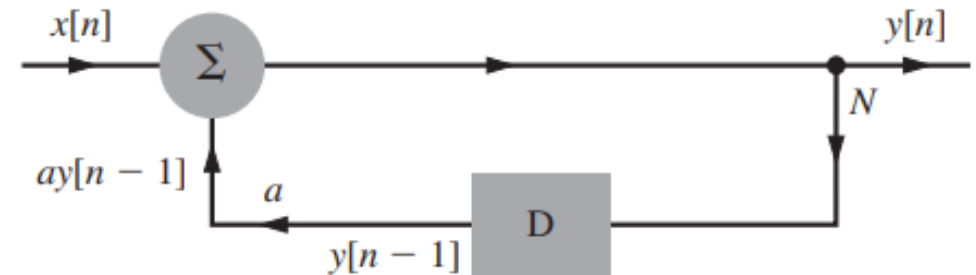


Figure 3.14 Realization of the savings account system.

DT LTI System Model:

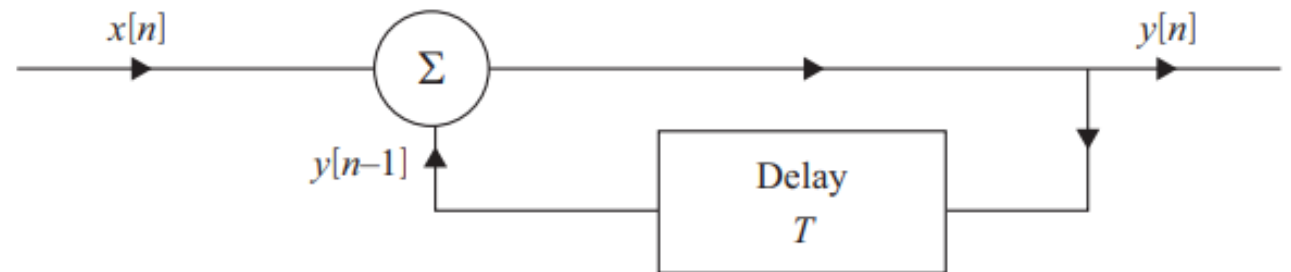
3-4: EXAMPLES OF DISCRETE-TIME SYSTEMS

3.4-1 A cash register output $y[n]$ represents the total cost of n items rung up by a cashier. The input $x[n]$ is the cost of the n th item.

- (a) Write the difference equation relating $y[n]$ to $x[n]$.
- (b) Realize this system using a time-delay element.

The signals are inherently discrete-time

$$y[n] - y[n - 1] = x[n]$$



DT LTI System Model:

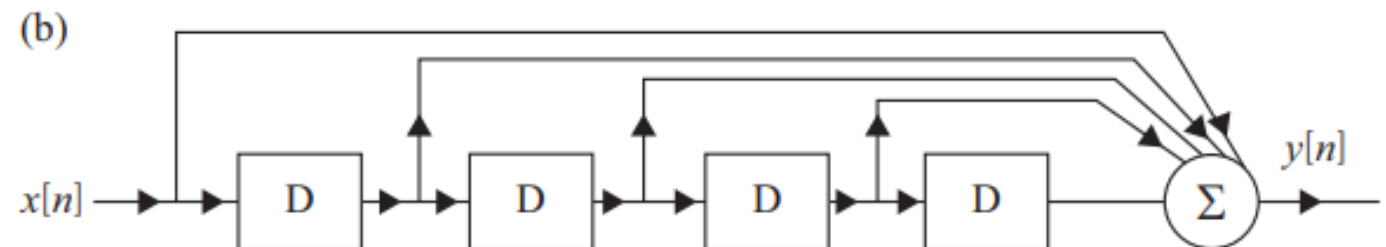
3-4: EXAMPLES OF DISCRETE-TIME SYSTEMS

3.4-3 A moving average is used to detect a trend of a rapidly fluctuating variable, such as the stock market average. A variable may fluctuate (up and down) daily, masking its long-term (secular) trend. We can discern the long-term trend by smoothing or averaging the past N values of the variable. For the stock market average, we may consider a 5-day moving average $y[n]$ to be the mean of the past 5 days' market closing values $x[n], x[n-1], \dots, x[n-4]$.

- (a) Write the difference equation relating $y[n]$ to the input $x[n]$.
- (b) Use time-delay elements to realize the 5-day moving-average filter.

The signals are inherently discrete-time

$$y[n] = x[n] + x[n-1] + x[n-2] + x[n-3] + x[n-4]$$



D represents unit delay

DT LTI System Model:

3-4: EXAMPLES OF DISCRETE-TIME SYSTEMS

EXAMPLE 3.8 Digital Differentiator

$$y(t) = \frac{dx(t)}{dt}$$

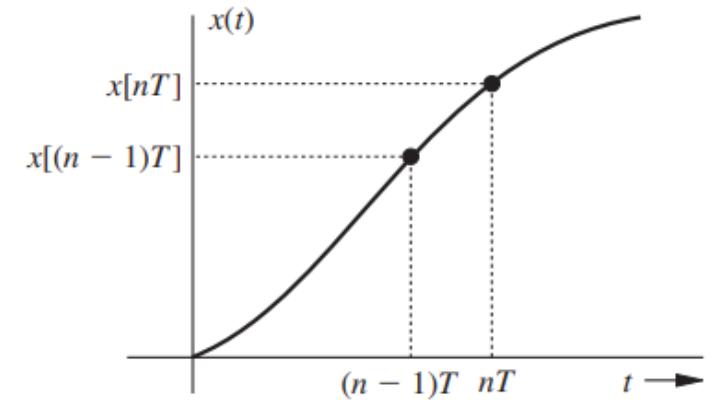
$$y(nT) = \left. \frac{dx(t)}{dt} \right|_{t=nT} = \lim_{T \rightarrow 0} \frac{1}{T} [x(nT) - x[(n-1)T]]$$

$$y[n] = \lim_{T \rightarrow 0} \frac{1}{T} \{x[n] - x[n-1]\}$$

$$y[n] = \frac{1}{T} \{x[n] - x[n-1]\}$$

Difference equations are the DT counterpart of differential equations.

A discrete-time system processes a continuous-time signal

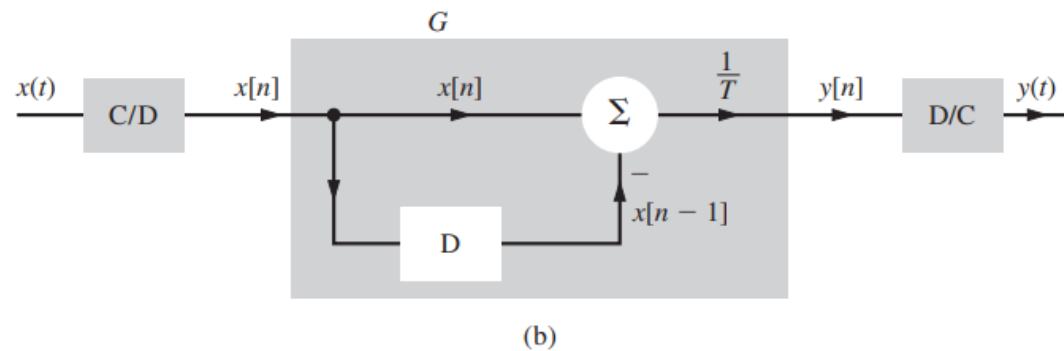


DT LTI System Model:

3-4: EXAMPLES OF DISCRETE-TIME SYSTEMS

EXAMPLE 3.8 Digital Differentiator

A discrete-time system processes a continuous-time signal



$$y[n] = \frac{1}{T} \{x[n] - x[n-1]\}$$

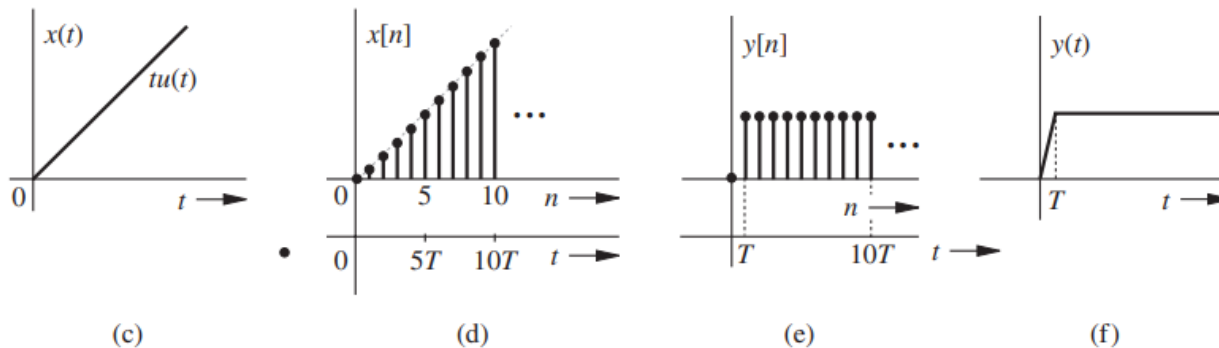


Figure 3.16 Digital differentiator and its realization.

DT LTI System Model:

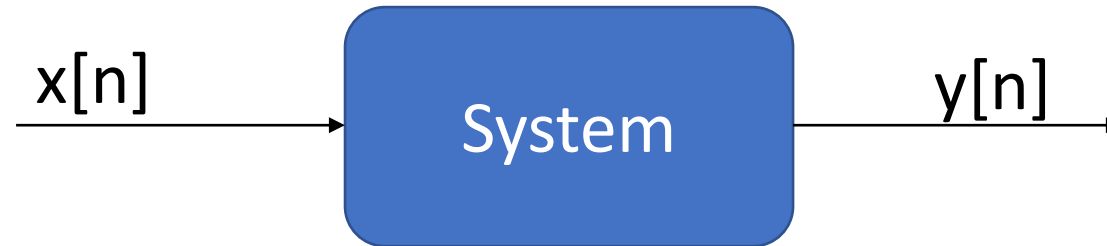
3-5: DISCRETE-TIME SYSTEMS EQUATIONS

Current input : $x[n]$

Past Inputs: $x[n-1]$, $x[n-2]$, $x[n-3]$,..... $x[n-N]$

Current output : $y[n]$

Past outputs: $y[n-1]$, $y[n-2]$, $y[n-3]$,..... $y[n-N]$



We can model the output of a DT LTI system as a linear combination of the current input, the past input, and past output values.

$$y[n] = -a_1y[n-1] - a_2y[n-2] - \dots - a_Ny[n-N] \\ + b_0x[n] + b_1x[n-1] + \dots + b_Nx[n-N]$$

DT LTI System Model:

3-5: DISCRETE-TIME SYSTEMS EQUATIONS

$$y[n] = -a_1y[n-1] - a_2y[n-2] - \dots - a_Ny[n-N] \\ + b_0x[n] + b_1x[n-1] + \dots + b_Nx[n-N]$$

This equation is called a [Linear Constant-Coefficient Difference Equation \(LCCDE\)](#), here

- The word difference is due to the delayed version of input $x[n]$ and output $y[n]$
- The coefficients $\{a_N, b_N\}$ are constants and not time-dependent due to the time-invariance nature of the system i.e. DT LTI

$$y[n] = \sum_{k=0}^N b_k x[n-k] - \sum_{l=1}^N a_l y[n-l]$$



DT LTI System Model:

3-5: DISCRETE-TIME SYSTEMS EQUATIONS

$$y[n] = -a_1y[n-1] - a_2y[n-2] - \dots - a_Ny[n-N] \\ + b_0x[n] + b_1x[n-1] + \dots + b_Nx[n-N]$$

$y[n-1], y[n-2], \dots$ *These terms give the system memory and create feedback.*

$x[n], x[n-1], x[n-2], \dots$ *These terms represent how the input drives the system.*

Example: FIR (Finite Impulse Response) system

$$y[n] = 0.5x[n] + 0.5x[n-1]$$

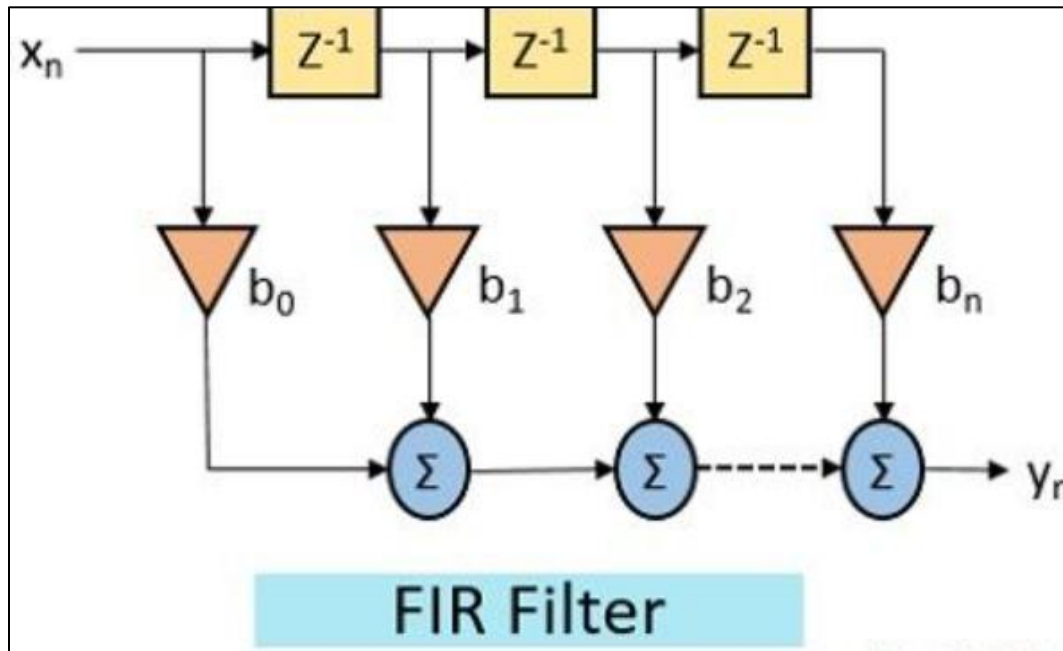
Example: IIR (Infinite Impulse Response) system

$$y[n] - 0.5y[n-1] + 0.2y[n-2] = x[n] + 0.3x[n-1]$$

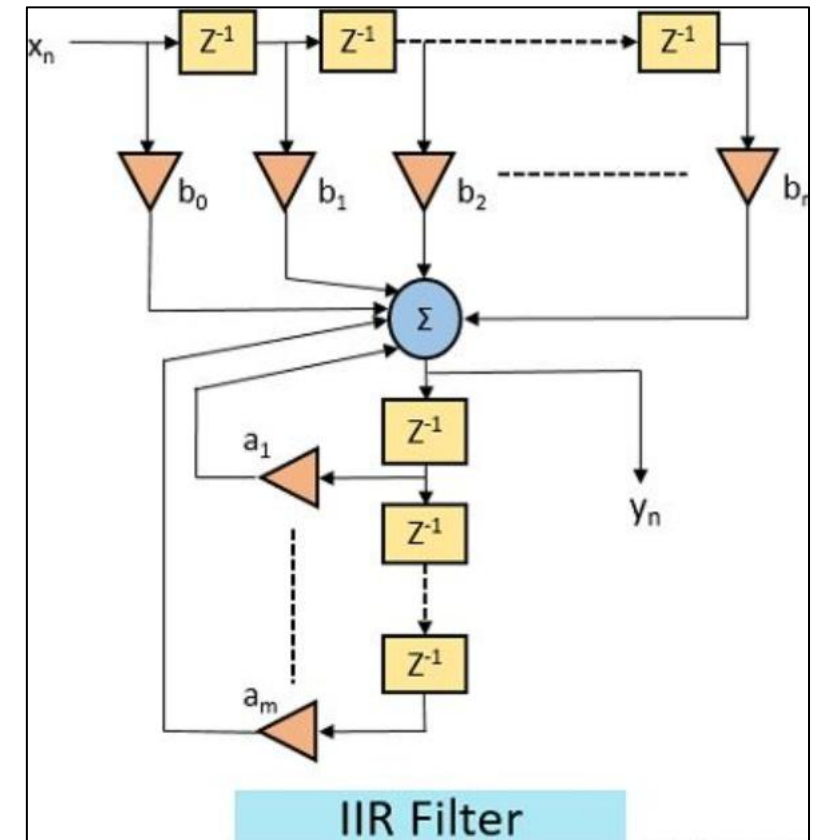
DT LTI System Model:

Example: FIR (Finite Impulse Response) system

Here delay $x[n-1]$ is modeled as Z^{-1}



Example: IIR (Infinite Impulse Response) system



Difference Equation solution
Time-domain Recursive/ iterative method

DT LTI System Model:

3-5: DISCRETE-TIME SYSTEMS EQUATIONS

Recursive (iterative) solution of a difference equation?

EXAMPLE 3.11 Iterative Solution to a First-Order Difference Equation

Solve iteratively

$$y[n] - 0.5y[n-1] = x[n]$$

with initial condition $y[-1] = 16$ and causal input $x[n] = n^2u[n]$. This equation can be expressed as

$$y[n] = 0.5y[n-1] + x[n] \quad (3.18)$$

If we set $n = 0$ in Eq. (3.18), we obtain

$$\begin{aligned} y[0] &= 0.5y[-1] + x[0] \\ &= 0.5(16) + 0 = 8 \end{aligned}$$

Now, setting $n = 1$ in Eq. (3.18) and using the value $y[0] = 8$ (computed in the first step) and $x[1] = (1)^2 = 1$, we obtain

$$y[1] = 0.5(8) + (1)^2 = 5$$

Next, setting $n = 2$ in Eq. (3.18) and using the value $y[1] = 5$ (computed in the previous step) and $x[2] = (2)^2$, we obtain

$$y[2] = 0.5(5) + (2)^2 = 6.5$$

Continuing in this way iteratively, we obtain

$$y[3] = 0.5(6.5) + (3)^2 = 12.25$$

$$y[4] = 0.5(12.25) + (4)^2 = 22.125$$

$\vdots$

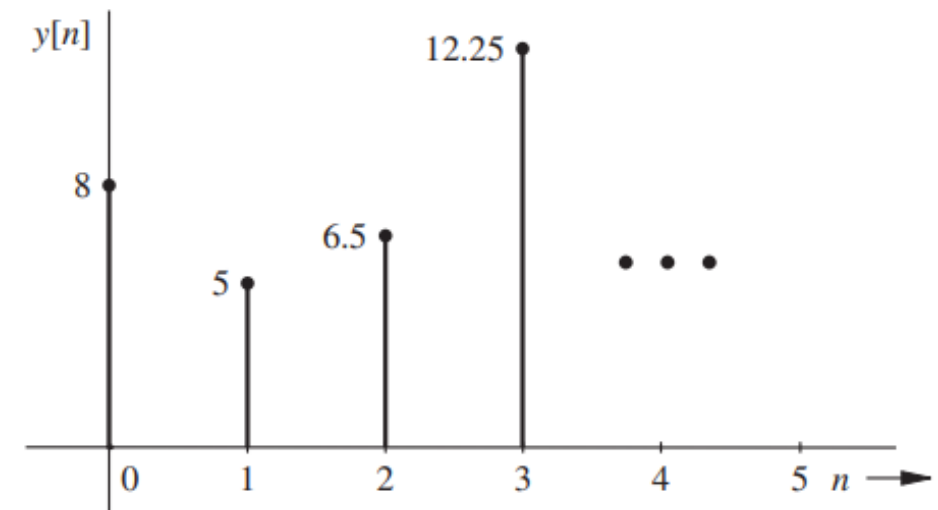


Figure 3.17 Iterative solution of a difference equation.



DT LTI System Model:

3-5: DISCRETE-TIME SYSTEMS EQUATIONS

Recursive (iterative) solution of a difference equation?

EXAMPLE 3.12 Iterative Solution to a Second-Order Difference Equation

Solve iteratively

$$y[n+2] - y[n+1] + 0.24y[n] = x[n+2] - 2x[n+1]$$

with initial conditions $y[-1] = 2$, $y[-2] = 1$ and a causal input $x[n] = nu[n]$. The system equation can be expressed as

$$y[n+2] = y[n+1] - 0.24y[n] + x[n+2] - 2x[n+1] \quad (3.19)$$

Setting $n = -2$ in Eq. (3.19) and then substituting $y[-1] = 2$, $y[-2] = 1$, $x[0] = x[-1] = 0$, we obtain

$$y[0] = 2 - 0.24(1) + 0 - 0 = 1.76$$

Setting $n = -1$ in Eq. (3.19) and then substituting $y[0] = 1.76$, $y[-1] = 2$, $x[1] = 1$, $x[0] = 0$, we obtain

$$y[1] = 1.76 - 0.24(2) + 1 - 0 = 2.28$$

Setting $n = 0$ in Eq. (3.19) and then substituting $y[0] = 1.76$, $y[1] = 2.28$, $x[2] = 2$, and $x[1] = 1$ yield

$$y[2] = 2.28 - 0.24(1.76) + 2 - 2(1) = 1.8576$$

MATLAB to compute other values of n

Difference Equation solution
Time-domain Recursive/ iterative method
Zero input & Zero state Response



DT LTI System Model:

- As we have seen, in a discrete-time system, the output $y[n]$ can be analyzed directly in the time domain using the difference equation of the system.
- The total response of a discrete-time system is divided into two components:
 - I. The system has initial conditions (previous outputs or memory elements).
 - II. The input signal $x[n]$ is applied

$$y[n] = y_{zi}[n] + y_{zs}[n]$$

where

- $y_{zi}[n]$ = Zero-Input Response (ZIR)
- $y_{zs}[n]$ = Zero-State Response (ZSR)

DT LTI System Model:

Zero-Input Response (ZIR):

- It represents the response caused by the previous outputs or memory elements.
- The zero-input response is the output of the system when the **input is set to zero**

$$y[n] = \sum_{k=0}^N b_k x[n-k] - \sum_{l=1}^N a_l y[n-l]$$

(A red arrow points from the word "ZERO" to the input term $b_k x[n-k]$ in the equation above.)

$$y[n] = -a_1 y[n-1] - a_2 y[n-2] - \dots - a_N y[n-N] \\ + b_0 x[n] + b_1 x[n-1] + \dots + b_N x[n-N]$$

(A red arrow points from the word "ZERO" to the input term $b_1 x[n-1]$ in the equation above.)



DT LTI System Model:

Zero-Input Response (ZIR):

Example:

$$y[n] + a_1y[n - 1] + a_2y[n - 2] = x[n]$$

For ZIR $x[n] = 0$

$$y[n] + a_1y[n - 1] + a_2y[n - 2] = 0$$

Solving this homogeneous difference equation gives the “Natural response”.



DT LTI System Model:

Zero-State Response (ZSR)

The zero-state response is the output when the input is applied, while **all initial conditions are zero**.

$$y[n] = \sum_{k=0}^N b_k x[n-k] - \sum_{l=1}^N a_l y[n-l]$$

Diagram illustrating the Zero-State Response (ZSR) equation. Red arrows point from the word "ZERO" to the input term $x[n-k]$ and the output term $y[n-l]$, indicating that both are zero in this context.

$$y[n] = -a_1 y[n-1] - a_2 y[n-2] - \dots - a_N y[n-N] + b_0 x[n] + b_1 x[n-1] + \dots + b_N x[n-N]$$

Diagram illustrating the Zero-State Response (ZSR) equation. Red arrows point from the word "ZERO" to the output terms $y[n-1]$ and $y[n-2]$, indicating that these terms are zero in this context.

DT LTI System Model:

Total Response = ZIR + ZSR

Example:

At $t < 0$ (Open circuit)

V_{dc} (input supply) not connected with RC circuit- **ZIR**

The voltage across the capacitor depends upon its charged voltage V_0

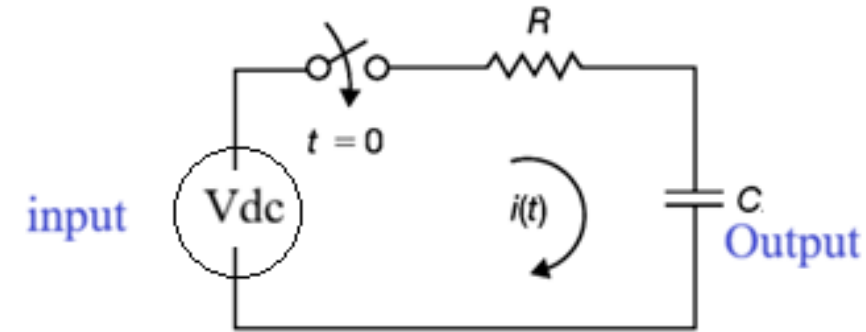
Output voltage = V_0

At $t = 0$ (Circuit closed)

Assume there is no charge stored at the capacitor (zero initial state) – **ZSR**

The voltage across the capacitor starts to build depend upon the time constant (RC), and finally reaches a steady state.

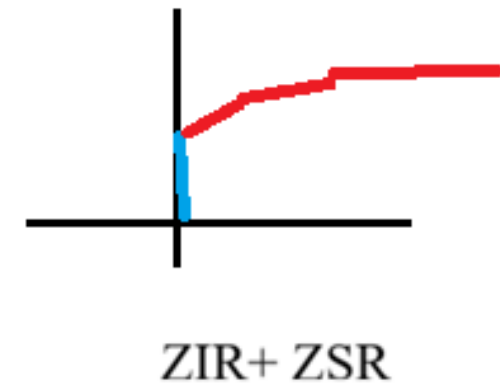
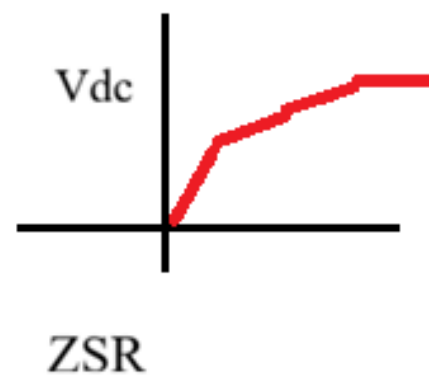
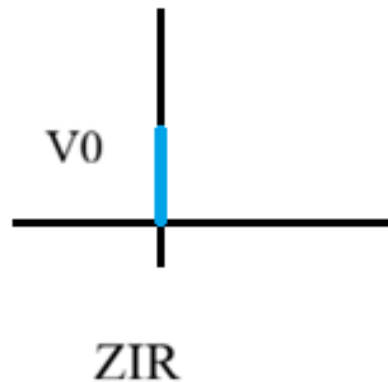
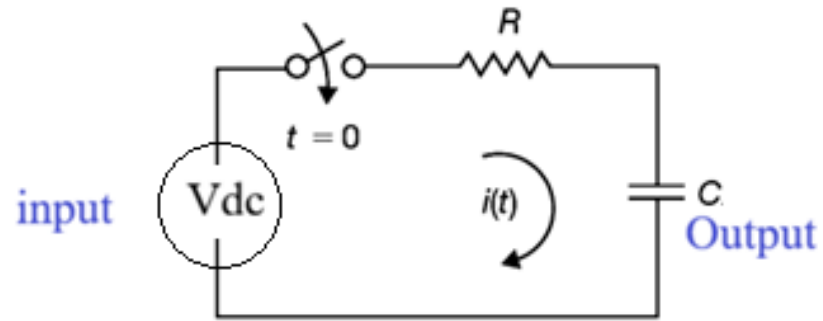
Output voltage = V_{dc}



DT LTI System Model:

Total Response = ZIR + ZSR

Example





Z Transform:

Example # 02

Total Response of System:

$$y[n] = 3x[n] + 0.8y[n - 1]$$

$$x[n] = \delta[n]$$

$$y[-1] = 10$$

$$y[n] = 3\delta[n] + 0.8y[n - 1]$$

$$y[0] = 3x[0] + 0.8y[-1] = 3(1) + 0.8(10) = 3 + 8 = 11$$

$$y[1] = 3x[1] + 0.8y[0] = 3(0) + 0.8(11) = 8.8$$

$$y[2] = 3x[2] + 0.8y[1] = 0 + 0.8(8.8) = 7.04$$

$$y[n] = 11 \cdot (0.8)^n, \quad n \geq 0$$

DT LTI System Model:

Example # 02

Zero-Input Response (ZIR)

$$y[n] = 3x[n] + 0.8y[n-1]$$

$$y[-1] = 10 \quad x[n] = \delta[n]$$

$$x[n] = 0$$

$$y[n] = \cancel{3x[n]} + 0.8y[n-1]$$

$$y[n] = 0.8y[n-1]$$

$$y[0] = 0.8y[-1] = 0.8(10) = 8$$

$$y[1] = 0.8y[0] = 0.8(8) = 6.4$$

$$y[2] = 0.8y[1] = 0.8(6.4) = 5.12$$

$$y[3] = 0.8y[2] = 0.8(5.12) = 4.096$$

$$y_{zi}[n] = 10(0.8)^{n+1}$$

DT LTI System Model:

Example # 02

Zero-State Response (ZSR)

$$y[n] = 3x[n] + 0.8y[n-1]$$

$$y[-1] = 10 \quad x[n] = \delta[n]$$

$$y[-1] = 0$$

$$y[n] = 3x[n] + 0.8y[n-1]$$

$$y[n] = 3x[n] + 0.8y[n-1]$$

$$y[0] = 3x[0] + 0.8y[-1] = 3(1) + 0.8(0) = 3$$

$$y[1] = 3x[1] + 0.8y[0] = 3(0) + 0.8(3) = 2.4$$

$$y[2] = 3x[2] + 0.8y[1] = 0 + 0.8(2.4) = 1.92$$

$$y[3] = 3x[3] + 0.8y[2] = 0 + 0.8(1.92) = 1.536$$

$$y_{zs}[n] = 3(0.8)^n u[n]$$

DT LTI System Model:

Example # 02

$$y[n] = 3x[n] + 0.8y[n-1]$$

$$y[-1] = 10 \quad x[n] = \delta[n]$$

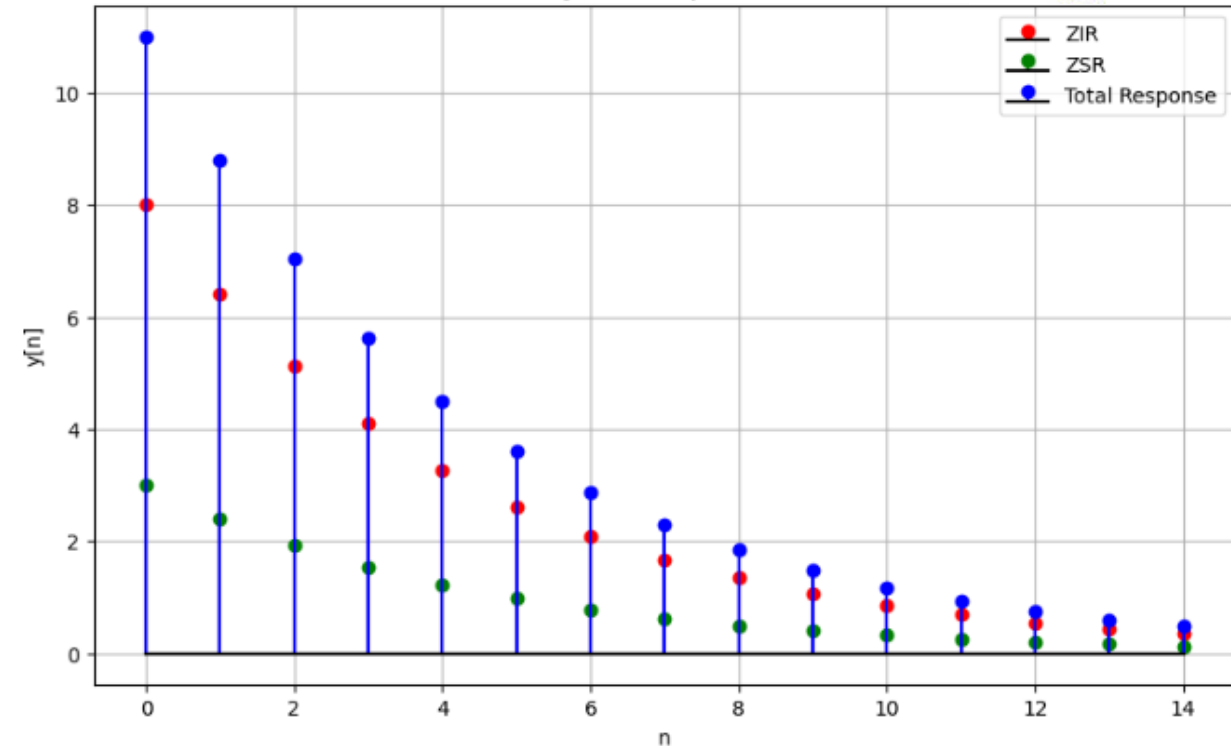
Total Response = ZIR + ZSR

$$y[n] = y_{zi}[n] + y_{zs}[n]$$

$$y_{zi}[n] = 10(0.8)^{n+1}$$

$$y_{zs}[n] = 3(0.8)^n u[n]$$

$$\begin{aligned} y[n] &= y_{zi}[n] + y_{zs}[n] \\ &= 10(0.8)^{n+1} + 3(0.8)^n \\ &= (0.8)^n (3 + 10 \cdot 0.8) \\ &= (0.8)^n (3 + 8) \\ &= 11(0.8)^n, \quad n \geq 0 \end{aligned}$$



Difference Equation solution

Frequency-domain Z Transform method



Learning outcome of today's session:

Z- Transform solution of Difference Equation – 5.3

- Z Transform – Systems Perspective
- DT System Modeling from Difference Equations
- Solution of Difference Equations:
 - Time- domain Recursive/iterative method
 - DT System total response – Zero input & Zero State response
 - Frequency-domain Z transform method

DT LTI System Model:

Current input : $x[n]$
 Past Inputs: $x[n-1]$, $x[n-2]$, $x[n-3]$,..... $x[n-N]$

Current output : $y[n]$
 Past outputs: $y[n-1]$, $y[n-2]$, $y[n-3]$,..... $y[n-N]$



We can model the output of a DT LTI system as a linear combination of the current input, the past input, and past output values.

$$\begin{aligned}
 y[n] = & -a_1y[n-1] - a_2y[n-2] - \dots - a_Ny[n-N] \\
 & + b_0x[n] + b_1x[n-1] + \dots + b_Nx[n-N]
 \end{aligned}$$

DT LTI System Model:

$$y[n] = -a_1y[n-1] - a_2y[n-2] - \dots - a_Ny[n-N] \\ + b_0x[n] + b_1x[n-1] + \dots + b_Nx[n-N]$$

This equation is called a [Linear Constant-Coefficient Difference Equation \(LCCDE\)](#), here

- The word difference is due to the delayed version of input $x[n]$ and output $y[n]$
- The coefficients $\{a_N, b_N\}$ are constants and not time- dependent due to the time-invariance nature of the system i.e. DT LTI

$$y[n] = \sum_{k=0}^N b_k x[n-k] - \sum_{l=1}^N a_l y[n-l]$$

DT LTI System Model:

Zero-Input Response (ZIR):

- It represents the response caused by the previous outputs or memory elements.
- The zero-input response is the output of the system when the **input is set to zero**

$$y[n] = \sum_{k=0}^N b_k x[n-k] - \sum_{l=1}^N a_l y[n-l]$$

A red arrow points from the word "ZERO" above the equation to the $x[n-k]$ term in the first sum.

$$y[n] = -a_1 y[n-1] - a_2 y[n-2] - \dots - a_N y[n-N] \\ + b_0 x[n] + b_1 x[n-1] + \dots + b_N x[n-N]$$

A red arrow points from the word "ZERO" below the equation to the $b_1 x[n-1]$ term in the second sum.

DT LTI System Model:

Zero-State Response (ZSR)

The zero-state response is the output when the input is applied, while **all initial conditions are zero**.

$$y[n] = \sum_{k=0}^N b_k x[n-k] - \sum_{l=1}^N a_l y[n-l]$$

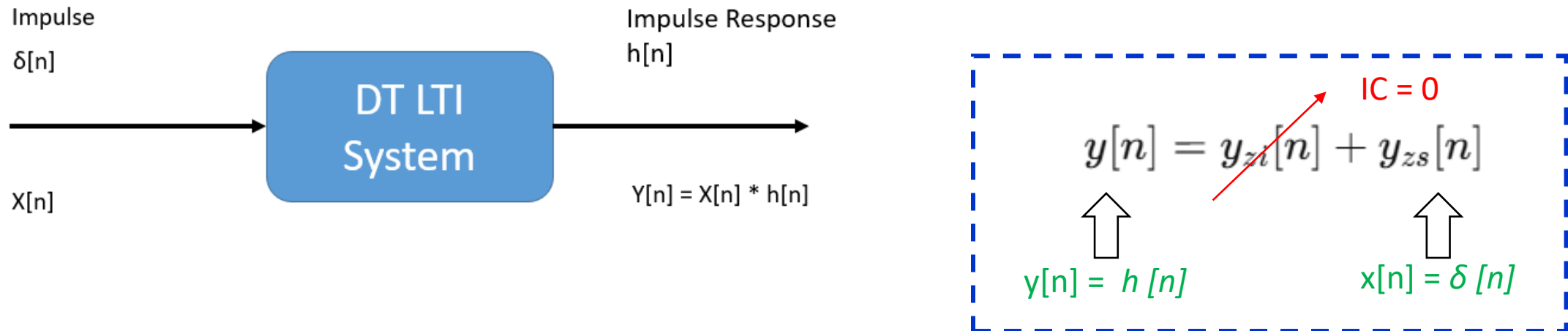
ZERO

$$y[n] = -a_1 y[n-1] - a_2 y[n-2] - \dots - a_N y[n-N] + b_0 x[n] + b_1 x[n-1] + \dots + b_N x[n-N]$$

ZERO

DT LTI System Model:

Important insight: In the case of the impulse response of an LTI system $h[n]$, we considered the input as a delta function, and we did not consider any initial conditions.



- So, we can say that the $h[n]$ is the Zero-state response (ZSR) of the LTI system with a specific input signal, i.e., $\delta[n]$
- Also, we can say that impulse response $h[n]$ is a special case of systems total response { no initial condition- ZIR, input $x[n] = \delta[n]$ }



DT LTI System Model:

Are there other (possibly better) ways to solve and analyze difference equations of DT-LTI systems instead of the time-domain iterative/recursive method?

YES- Frequency Transforms

Need for Transforms in Signal & Systems Analysis:



- Transforms provide frequency-domain insight
- Transforms help to attain a compact mathematical description of Signals & Systems using complex exponentials.
 - Transforms allow decomposition of the signal into complex sinusoids ($e^{j\omega}$)
 - Transforms also allow system representation in exponential form ($e^{\sigma+j\omega}$ or $re^{j\omega}$)
- Transforms convert convolution in the time domain into multiplication in the frequency domain
- Transform converts differential (continuous time)/ difference equations (discrete time) into algebraic equations.





Z Transform:

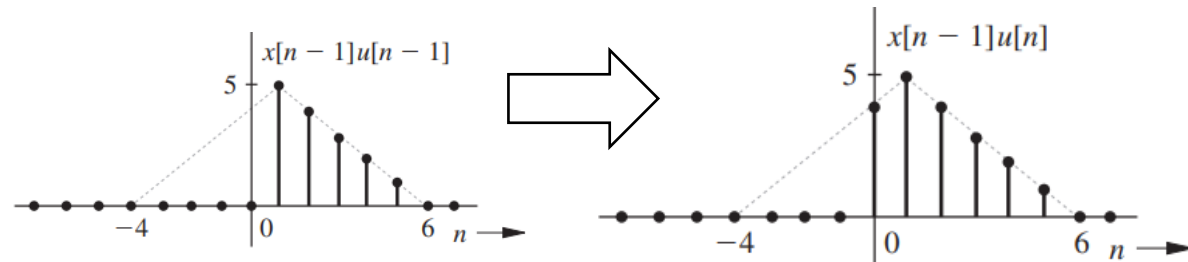
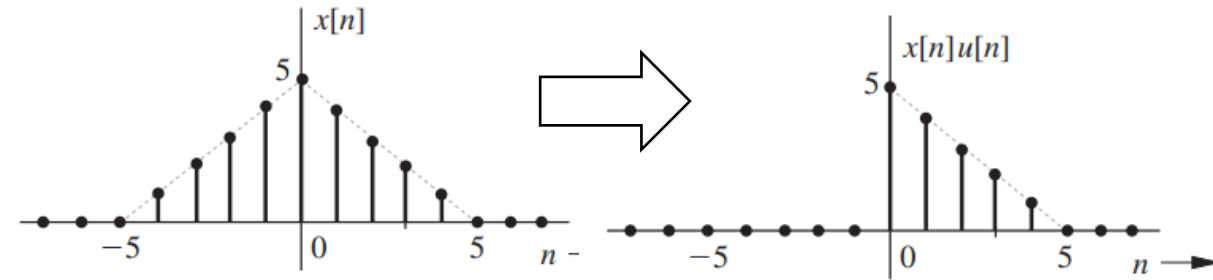
- LTI systems can be modeled as a linear combination of past inputs and outputs. This approach for the DT LTI system leads to difference equations.
- Z- transform converts a difference equation into simple algebraic equations, avoiding iterative solution.
- Therefore, the Z-transform helps to solve these difference equations elegantly.

Z Transform:

Operation	$x[n]$	$X[z]$
Addition	$x_1[n] + x_2[n]$	$X_1[z] + X_2[z]$
Scalar multiplication	$ax[n]$	$aX[z]$
Right shifting	$x[n - m]u[n - m]$	$\frac{1}{z^m} X[z]$
	$x[n - m]u[n]$	$\frac{1}{z^m} X[z] + \frac{1}{z^m} \sum_{n=1}^m x[-n]z^n$
	$x[n - 1]u[n]$	$\frac{1}{z} X[z] + x[-1]$
	$x[n - 2]u[n]$	$\frac{1}{z^2} X[z] + \frac{1}{z} x[-1] + x[-2]$
	$x[n - 3]u[n]$	$\frac{1}{z^3} X[z] + \frac{1}{z^2} x[-1] + \frac{1}{z} x[-2] + x[-3]$
Left shifting	$x[n + m]u[n]$	$z^m X[z] - z^m \sum_{n=0}^{m-1} x[n]z^{-n}$
	$x[n + 1]u[n]$	$zX[z] - zx[0]$
	$x[n + 2]u[n]$	$z^2 X[z] - z^2 x[0] - zx[1]$
	$x[n + 3]u[n]$	$z^3 X[z] - z^3 x[0] - z^2 x[1] - zx[2]$

Z Transform:

Right Shifting property



RIGHT SHIFT (DELAY)

If

$$x[n]u[n] \iff X[z]$$

then

$$x[n-1]u[n-1] \iff \frac{1}{z}X[z]$$

In general,

$$x[n-m]u[n-m] \iff \frac{1}{z^m}X[z]$$

Moreover,

$$x[n-1]u[n] \iff \frac{1}{z}X[z] + x[-1]$$

Repeated application of this property yields

$$x[n-2]u[n] \iff \frac{1}{z} \left[\frac{1}{z}X[z] + x[-1] \right] + x[-2] = \frac{1}{z^2}X[z] + \frac{1}{z}x[-1] + x[-2]$$

In general, for integer value of m ,

$$x[n-m]u[n] \iff z^{-m}X[z] + z^{-m} \sum_{n=1}^m x[-n]z^n$$



Z Transform:

Example: 02 - Again

Solve iteratively in the time domain
(without Z transform)

$$y[n] = 3\delta[n] + 0.8y[n - 1]$$

$$y[0] = 3x[0] + 0.8y[-1] = 3(1) + 0.8(10) = 3 + 8 = 11$$

$$y[1] = 3x[1] + 0.8y[0] = 3(0) + 0.8(11) = 8.8$$

$$y[2] = 3x[2] + 0.8y[1] = 0 + 0.8(8.8) = 7.04$$

$$y[n] = 11 \cdot (0.8)^n, \quad n \geq 0$$

$$y[n] = 3x[n] + 0.8y[n - 1]$$

$$x[n] = \delta[n]$$

$$y[-1] = 10$$



Z Transform:

Example: 02 $y[n] = 3x[n] + 0.8y[n - 1]$

Solve using the Z transform

$$Z\{y[n]\} = Y(z), \quad Z\{x[n]\} = X(z)$$

$$x[n] = \delta[n]$$

$$y[-1] = 10$$

$$Z\{y[n - 1]\} = z^{-1}Y(z) + y[-1]$$

$$Y(z) = 3X(z) + 0.8(z^{-1}Y(z) + y[-1])$$

Substitute $X(z) = Z\{\delta[n]\} = 1$ and $y[-1] = 10$:

$$Y(z) = 3 + 0.8z^{-1}Y(z) + 0.8(10)$$

$$Y(z) = 3 + 0.8z^{-1}Y(z) + 8$$

$$Y(z) = 11 + 0.8z^{-1}Y(z)$$

$$Y(z) - 0.8z^{-1}Y(z) = 11$$

$$Y(z)(1 - 0.8z^{-1}) = 11$$

Right Shifting property

$$x[n - 1]u[n] \longleftrightarrow \frac{1}{z}X[z] + x[-1]$$

$$Y(z) = \frac{11}{1 - 0.8z^{-1}}$$

$$Y(Z) = \frac{11Z}{Z - 0.8}$$

Inverse Z Transform

$$y[n] = 11(0.8)^n u[n], \quad n \geq 0$$

$$\gamma^n u[n] \longleftrightarrow \frac{z}{z - \gamma}$$



Z Transform:

Example: 02

Solve using the Z transform

$$Z\{y[n]\} = Y(z), \quad Z\{x[n]\} = X(z)$$

$$Z\{y[n-1]\} = z^{-1}Y(z) + y[-1]$$

$$Y(z) = 3X(z) + 0.8(z^{-1}Y(z) + y[-1])$$

Substitute $X(z) = Z\{\delta[n]\} = 1$ and $y[-1] = 10$:

$$Y(z) = 3 + 0.8z^{-1}Y(z) + 0.8(10)$$

$$Y(z) = 3 + 0.8z^{-1}Y(z) + 8$$

$$y[n] = 3x[n] + 0.8y[n-1]$$

$$x[n] = \delta[n]$$

$$y[-1] = 10$$

ZIR

$$Y(z) = \frac{8}{1 - 0.8z^{-1}}$$

$$Y(Z) = \frac{8Z}{Z - 0.8}$$

Inverse Z Transform

$$y[n] = 8 (0.8)^n u[n],$$

ZSR

$$Y(z) = \frac{11}{1 - 0.8z^{-1}}$$

$$Y(Z) = \frac{3Z}{Z - 0.8}$$

Inverse Z Transform

$$y[n] = 3 (0.8)^n u[n], \quad n \geq 0$$



Z Transform:

Class Participation Quiz

Example: 03

$$y[n] - 5y[n - 1] + 6y[n - 2] = 3x[n - 1] + 5x[n - 2]$$

$$x[n] = (2)^{-n}u[n]$$

$$y[-1] = 11/6, y[-2] = 37/36$$

Answer Key:

$$Y[z] = \frac{z(3z^2 - 9.5z + 10.5)}{(z - 0.5)(z^2 - 5z + 6)}$$

Hint:

$$x[n] = (2)^{-n}u[n] = (2^{-1})^n u[n] = (0.5)^n u[n] \iff \frac{z}{z - 0.5}$$

Operation	$x[n]$	$X[z]$
Addition	$x_1[n] + x_2[n]$	$X_1[z] + X_2[z]$
Scalar multiplication	$ax[n]$	$aX[z]$
Right shifting	$x[n - m]u[n - m]$	$\frac{1}{z^m}X[z]$
	$x[n - m]u[n]$	$\frac{1}{z^m}X[z] + \frac{1}{z^m} \sum_{n=1}^m x[-n]z^n$
	$x[n - 1]u[n]$	$\frac{1}{z}X[z] + x[-1]$
	$x[n - 2]u[n]$	$\frac{1}{z^2}X[z] + \frac{1}{z}x[-1] + x[-2]$
	$x[n - 3]u[n]$	$\frac{1}{z^3}X[z] + \frac{1}{z^2}x[-1] + \frac{1}{z}x[-2] + x[-3]$
Left shifting	$x[n + m]u[n]$	$z^mX[z] - z^m \sum_{n=0}^{m-1} x[n]z^{-n}$
	$x[n + 1]u[n]$	$zX[z] - zx[0]$
	$x[n + 2]u[n]$	$z^2X[z] - z^2x[0] - zx[1]$
	$x[n + 3]u[n]$	$z^3X[z] - z^3x[0] - z^2x[1] - zx[2]$

Z Transform:

Example: 03

Right Shifting property

$x[n-1]u[n]$	$\frac{1}{z}X[z] + x[-1]$
$x[n-2]u[n]$	$\frac{1}{z^2}X[z] + \frac{1}{z}x[-1] + x[-2]$
$x[n-3]u[n]$	$\frac{1}{z^3}X[z] + \frac{1}{z^2}x[-1] + \frac{1}{z}x[-2] + x[-3]$

$$y[n] - 5y[n-1] + 6y[n-2] = 3x[n-1] + 5x[n-2]$$

$$Y[z]$$

$$\frac{1}{z}Y[z] + y[-1]$$

$$\frac{1}{z^2}Y[z] + \frac{1}{z}y[-1] + y[-2]$$

$$\frac{1}{z}X[z] + x[-1]$$

$$\frac{1}{z^2}X[z] + \frac{1}{z}x[-1] + x[-2]$$

$$y[-1] = 11/6, y[-2] = 37/36$$

$$Y[z] - 5 \left[\frac{1}{z}Y[z] + \frac{11}{6} \right] + 6 \left[\frac{1}{z^2}Y[z] + \frac{1}{z} \cdot \frac{11}{6} + \frac{37}{36} \right] = 3 \left[\frac{1}{z}X[z] \right] + 5 \left[\frac{1}{z^2}X[z] \right]$$

Z Tx has converted the difference equation into an algebraic equation!



Z Transform:

Example: 03

$$x[n] = (2)^{-n}u[n].$$

$$y[-1] = 11/6, y[-2] = 37/36.$$

$$y[n] - 5y[n-1] + 6y[n-2] = 3x[n-1] + 5x[n-2]$$

$$Y[z] - 5 \left[\frac{1}{z} Y[z] + \frac{11}{6} \right] + 6 \left[\frac{1}{z^2} Y[z] + \frac{1}{z} \cdot \frac{11}{6} + \frac{37}{36} \right] = 3 \left[\frac{1}{z} X[z] \right] + 5 \left[\frac{1}{z^2} X[z] \right]$$

To solve the algebraic equation for $Y[z]$ we only need to find $X[z]$

$$x[n] = (2)^{-n}u[n] = (2^{-1})^n u[n] = (0.5)^n u[n] \iff \frac{z}{z-0.5}$$

Now, simplify the equation to get the final expression of $Y[z]$.



Z Transform:

Example: 03

$$Y[z] - 5 \left[\frac{1}{z} Y[z] + \frac{11}{6} \right] + 6 \left[\frac{1}{z^2} Y[z] + \frac{1}{z} \cdot \frac{11}{6} + \frac{37}{36} \right] = 3 \left[\frac{1}{z} X[z] \right] + 5 \left[\frac{1}{z^2} X[z] \right]$$

$$Y[z] - \frac{5}{z} Y[z] - \frac{55}{6} + \frac{6}{z^2} Y[z] + \frac{11}{z} + \frac{37}{6} = 3 \left[\frac{1}{z} X[z] \right] + 5 \left[\frac{1}{z^2} X[z] \right]$$

$$Y[z] \left(1 - \frac{5}{z} + \frac{6}{z^2} \right) + \frac{11}{z} - 3 = 3 \left[\frac{1}{z} X[z] \right] + 5 \left[\frac{1}{z^2} X[z] \right]$$

$$Y[z] \left(1 - \frac{5}{z} + \frac{6}{z^2} \right) = \frac{3}{z} X[z] + \frac{5}{z^2} X[z] - \frac{11}{z} + 3$$

$$Y[z](z^2 - 5z + 6) = 3zX[z] + 5X[z] - 11z + 3z^2$$



Z Transform:

Example: 03

$$Y[z](z^2 - 5z + 6) = 3zX[z] + 5X[z] - 11z + 3z^2$$

$$Y[z](z^2 - 5z + 6) = \frac{3z^2 + 5z}{z - 0.5} - 11z + 3z^2$$

$$Y[z](z^2 - 5z + 6) = \frac{3z^3 - 9.5z^2 + 10.5z}{z - 0.5}$$

$$Y[z] = \frac{3z^3 - 9.5z^2 + 10.5z}{(z - 0.5)(z^2 - 5z + 6)}$$

$$Y[z] = \frac{z(3z^2 - 9.5z + 10.5)}{(z - 0.5)(z^2 - 5z + 6)}$$

$$3zX[z] = \frac{3z^2}{z - 0.5}$$

$$5X[z] = \frac{5z}{z - 0.5}$$



Z Transform:

Example: 03

$$Y[z] = \frac{z(3z^2 - 9.5z + 10.5)}{(z - 0.5)(z^2 - 5z + 6)}$$

Partial fraction expansion:

$$Y[z] = \frac{26}{15} \left(\frac{z}{z - 0.5} \right) - \frac{7}{3} \left(\frac{z}{z - 2} \right) + \frac{18}{5} \left(\frac{z}{z - 3} \right)$$

$$y[n] = \left[\frac{26}{15} (0.5)^n - \frac{7}{3} (2)^n + \frac{18}{5} (3)^n \right] u[n]$$



Z Transform:

Example: 02

In the context of ZIR & ZSR

$$Y[z] - 5 \left[\frac{1}{z} Y[z] + \frac{11}{6} \right] + 6 \left[\frac{1}{z^2} Y[z] + \frac{1}{z} \cdot \frac{11}{6} + \frac{37}{36} \right] = 3 \left[\frac{1}{z} X[z] \right] + 5 \left[\frac{1}{z^2} X[z] \right]$$

$$Y[z] - \frac{5}{z} Y[z] - \frac{55}{6} + \frac{6}{z^2} Y[z] + \frac{11}{z} + \frac{37}{6} = 3 \left[\frac{1}{z} X[z] \right] + 5 \left[\frac{1}{z^2} X[z] \right]$$

$$Y[z] \left(1 - \frac{5}{z} + \frac{6}{z^2} \right) + \frac{11}{z} - 3 = 3 \left[\frac{1}{z} X[z] \right] + 5 \left[\frac{1}{z^2} X[z] \right]$$

$$X[z] = z / (z - 0.5)$$

$$\underbrace{\left(1 - \frac{5}{z} + \frac{6}{z^2} \right) Y[z] - \left(3 - \frac{11}{z} \right)}_{\text{IC terms}} = \underbrace{\frac{3}{z - 0.5} + \frac{5}{z(z - 0.5)}}_{\text{input terms}}$$

Z Transform:

Example: 02

In the context of ZIR & ZSR

Therefore,

$$\left(1 - \frac{5}{z} + \frac{6}{z^2}\right) Y[z] = \underbrace{\left(3 - \frac{11}{z}\right)}_{\text{IC terms}} + \underbrace{\frac{(3z+5)}{z(z-0.5)}}_{\text{input terms}}$$

Multiplying both sides by z^2 yields

$$(z^2 - 5z + 6)Y[z] = \underbrace{z(3z - 11)}_{\text{IC terms}} + \underbrace{\frac{z(3z+5)}{z-0.5}}_{\text{input terms}}$$

and

$$Y[z] = \underbrace{\frac{z(3z-11)}{z^2-5z+6}}_{\text{zero-input response}} + \underbrace{\frac{z(3z+5)}{(z-0.5)(z^2-5z+6)}}_{\text{zero-state response}}$$

We expand both terms on the right-hand side into modified partial fractions to yield

$$Y[z] = \underbrace{\left[5\left(\frac{z}{z-2}\right) - 2\left(\frac{z}{z-3}\right)\right]}_{\text{zero-input response}} + \underbrace{\left[\frac{26}{15}\left(\frac{z}{z-0.5}\right) - \frac{22}{3}\left(\frac{z}{z-2}\right) + \frac{28}{5}\left(\frac{z}{z-3}\right)\right]}_{\text{zero-state response}}$$

and

$$\begin{aligned} y[n] &= \underbrace{(5(2)^n - 2(3)^n)u[n]}_{\text{zero-input response}} + \underbrace{\left(\frac{26}{15}(0.5)^n - \frac{22}{3}(2)^n + \frac{28}{5}(3)^n\right)u[n]}_{\text{zero-state response}} \\ &= \left[-\frac{7}{3}(2)^n + \frac{18}{5}(3)^n + \frac{26}{15}(0.5)^n\right]u[n] \end{aligned}$$



Z Transform:

DRILL 5.10 z-Transform Solution of a Linear Difference Equation

Solve the following equation if the initial conditions $y[-1] = 2$, $y[-2] = 0$, and the input $x[n] = u[n]$:

$$y[n+2] - \frac{5}{6}y[n+1] + \frac{1}{6}y[n] = 5x[n+1] - x[n]$$

ANSWER

$$y[n] = \left[12 - 15\left(\frac{1}{2}\right)^n + \frac{14}{3}\left(\frac{1}{3}\right)^n \right] u[n]$$

DRILL 5.11 Difference Equation Solution Using $y[0]$, $y[1]$, $\dots$, $y[N-1]$

Solve the following equation if the auxiliary conditions are $y[0] = 1$, $y[1] = 2$, and the input $x[n] = u[n]$:

$$y[n] + 3y[n-1] + 2y[n-2] = x[n-1] + 3x[n-2]$$

ANSWER

$$y[n] = \left[\frac{2}{3} + 2(-1)^n - \frac{5}{3}(-2)^n \right] u[n]$$

Z Transform:

DRILL 5.12 Separating Zero-Input and Zero-State Responses

Solve

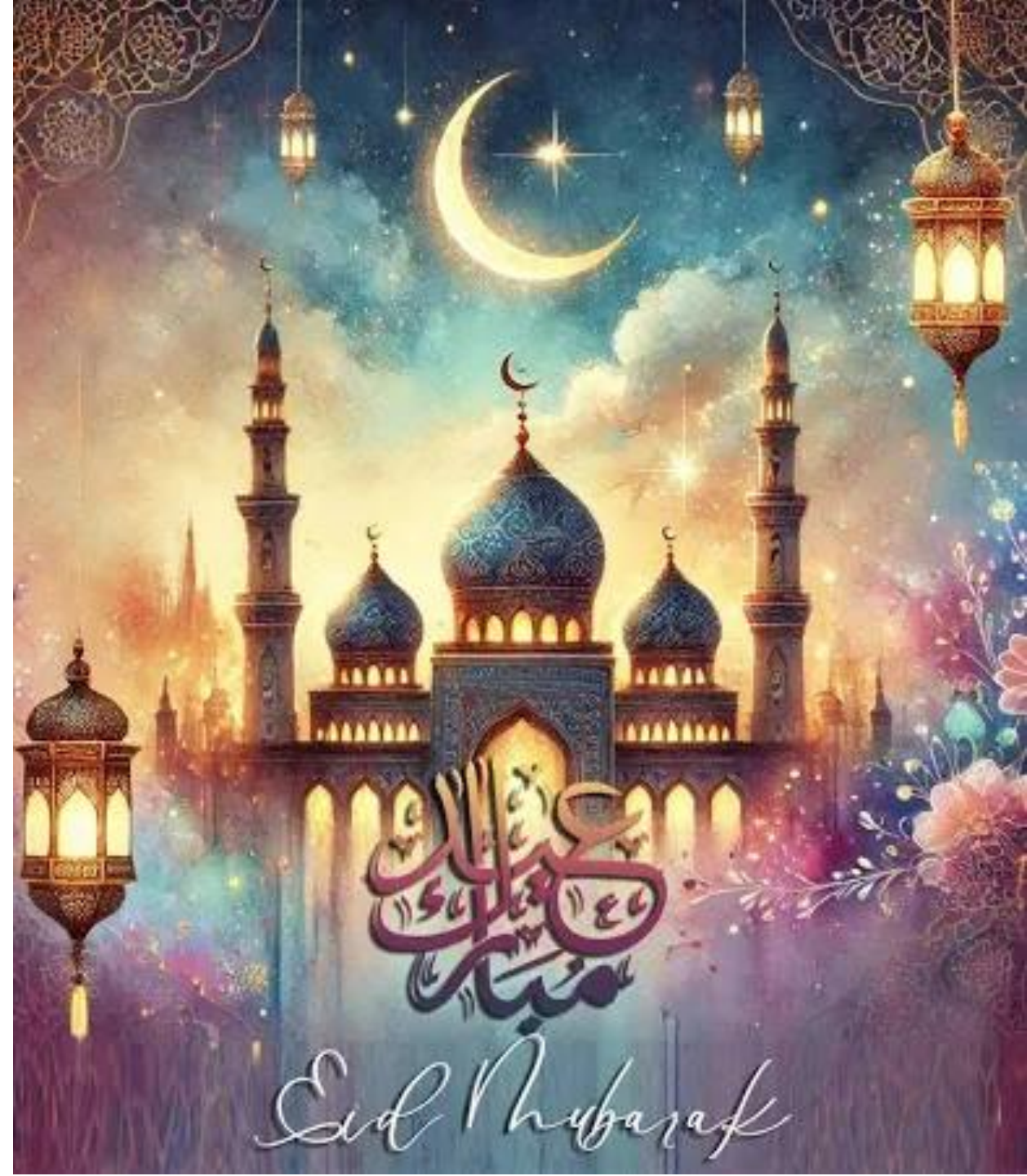
$$y[n+2] - \frac{5}{6}y[n+1] + \frac{1}{6}y[n] = 5x[n+1] - x[n]$$

if the initial conditions are $y[-1] = 2$, $y[-2] = 0$, and the input $x[n] = u[n]$. Separate the response into zero-input and zero-state responses.

ANSWER

$$\begin{aligned} y[n] &= \underbrace{\left(3\left(\frac{1}{2}\right)^n - \frac{4}{3}\left(\frac{1}{3}\right)^n\right)u[n]}_{\text{zero-input response}} + \underbrace{\left(12 - 18\left(\frac{1}{2}\right)^n + 6\left(\frac{1}{3}\right)^n\right)u[n]}_{\text{zero-state response}} \\ &= \left[12 - 15\left(\frac{1}{2}\right)^n + \frac{14}{3}\left(\frac{1}{3}\right)^n\right]u[n] \end{aligned}$$

Wishing you a blessed and joyful Eid celebration with your loved ones.





Thanks!